

Fiyinfoluwa David Osokoya

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Education

Air Force Institute of Technology

B.Eng. in Mechatronics Engineering

Kaduna, Nigeria

April 2021 – September 2025

- **GPA:** 4.86/5.00
- **Undergraduate Thesis:** Design and Development of an Automated Robotic System for Classification and Sorting using Machine Vision.
- **Relevant Coursework:** Machine Learning, Digital Signal Processing, Computer Vision, Control Systems, Artificial Intelligence.

Research Interests

Interested in developing human-interpretable generative models and representation learning techniques that bridge machine vision, signal processing, and intelligent systems with applications in neuroscience, brain-machine interfaces, and adaptive robotics in resource-constrained environments.

Research Experience

Air Force Institute of Technology (AFIT) Kaduna, Nigeria

Abuja, Nigeria

Research Student (MCT LAB)

October 2024 – Present

- Designed and prototyped a low-cost robotic sorting system integrating machine vision (OpenCV) and deep learning for real-time object classification and sorting on conveyor systems.
- Developed interpretable computer vision pipelines using sparse coding and representation learning to enhance robustness against noise and overlapping features in industrial visual streams.
- Co-authored publication (January 2026) incorporating peer-review feedback for robustness against industrial noise; secured Vice Chancellor's funding to scale the prototype.

Project Cage Technologies

Lagos, Nigeria

Research and Development Intern

May 2024 – October 2024

- Trained convolutional models and explored sparse autoencoder architectures for factory product inspection and defect detection in dynamic visual data streams.
- Applied software engineering principles to AI-enabled robotic systems, focusing on modular control architectures and maintainable component integration for adaptive behaviors.
- Built real-time obstacle-avoidance and human-robot interaction systems (<200ms response time) incorporating signal processing and adaptive control.

AFIT Digital Hub

Kaduna, Nigeria

Research Student

July 2023 – April 2024

- Developed a campus-wide social communication platform with integrated real-time chat, applying software engineering practices for AI-driven interaction systems.
- Developed Designed the platform using modular and reusable components to support scalability and maintain software ecosystem quality.
- Analyzed usage patterns and message flow data to evaluate system performance and improve reliability, aligning with mining software repositories and iterative software quality improvement.

Teaching Experience

WhiteSands Secondary School

Lagos, Nigeria

Python and Robotics Programming Tutor

July 2024 – September 2024

Project Cage Technologies

Lagos, Nigeria

Robotics Instructor

May 2024 – October 2024

Awards and Honors

Graduating Student Award, Overall Best Graduating Student for Department of Mechatronics

2025

Graduating Student Award, Overall Best Male Graduating Student for Department of Mechatronics

2025

Participant Award, Mechatronics Department Project Exhibition

2025

Excellent Performance Award , Foursquare Gospel Church	2025
Finalist , AFIT Digital Hub Hackathon	2023

Core Skills

Programming & Framework: Python, MATLAB, Git, JavaScript, React, Node.js
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face
Personal Skills: Leadership, Hyper-focus, Resilience, Growth Mindset, Teamwork

Selected Projects

DUNL Interactive Dashboard: Interactive Analytical Dashboard for Interpretable Deep Learning Ongoing
 • Building a research dashboard for visualizing training dynamics, learned representations (kernels), sparse codes, and signal reconstructions in time-series models.
 • Features include epoch time-travel, live training synchronization, and comparative analysis tools.

Summara: Video Upload, Transcript Extraction, and LLM-Prompting System
 • Designed a tool for uploading videos, extracting transcripts, and generating LLM-based prompts for video analysis and summarization, integrating natural language processing to support intelligent content processing.
 • Applied software engineering for AI to create modular pipelines that enhance developer assistance in multimedia applications, with potential extensions to robotic vision systems for real-time video interpretation in automation tasks.

RepoScope: LLM Powered GitHub Repository Mining and Analysis Tool 2025
 • Developed a software chat-bot style assistant that analyzes GitHub repositories using LLMs to summarize codebases, identify tech stacks and extract development patterns
 • Applied intelligent research discovery system using natural language processing to automatically parse, summarize, and Mining software repositories and software engineering for AI to analyze maintainability and software quality in open-source projects.

Undergraduate Thesis: Low-Cost Robotic Sorting System with Machine Vision 2025
 • Designed and prototyped an affordable robotic sorting system for small-scale manufacturers, integrating machine vision for product classification and sorting on a conveyor belt.
 • Co-authored publication (January 2026) incorporating peer-review feedback for robustness against industrial noise; secured Vice Chancellor's funding to scale the prototype.

Selected Presentations and Invited Talks

Designing Reliable Software Ecosystems for AI-Based Robotic Automation
Presenter, Departmental Final Year Project Exhibition, AFIT, July 2025.
The Need for Software Development/Robotics Education in Secondary School
Guest Speaker, Apostolic Faith Secondary School (AFSS), Lagos, September 2024.
360° Leadership: Leading from Every Level
Departmental Leadership Seminar, October 2023.

Professional Experience

NSMC Trading Academy <i>Software Engineer</i>	Kaduna, Nigeria (Remote) April 2025 – Present
- Built and deployed a scalable React-based learning platform with reusable UI components. - Collaborated Improved user experience through data-driven iteration, leading to a 10% increase in student engagement.	
FanMilk PLC <i>Mechatronics Intern</i>	Ibadan, Oyo State September 2023 – Jan 2024
- Explored modular manufacturing systems in beverage production, identifying limitations like manual sorting in West African operations. - Initiated concepts for affordable automation to reduce labor dependencies, focusing on machine vision and robotic intervention for improved efficiency.	

Taichi Multi-Investment Company
Software Engineer

- Developed modular financial dashboards to enhance data-driven insight and engagement.
- Extended internal software ecosystem components with real-time visualization and automated reporting workflows.
- Improved system performance, maintainability, and scalability through caching, responsive UI patterns, and optimization strategies.

Ibadan, Nigeria
Jan 2023 – September 2024

Leadership Experience

Air Force Institute of Technology
Course Representative

- Represented a class of 60+ students, coordinating coursework, schedules, and communication with faculty.
- Organized study groups and tutorial sessions, simplifying software engineering, AI, and system development concepts for peers.

Kaduna, Nigeria
September 2023 – September 2025

WhiteSands Robotics Club
Robotics Coach

- Trained and mentored a student robotics team that qualified for the PARC International Robotics Competition, earning commendation for effective leadership and technical guidance.
- Guided 20 students through project development, teamwork, and technical problem-solving, significantly boosting their confidence and competitive readiness.

Lagos, Nigeria
May 2024 – September 2024